# 10. $\sigma$-algebras, measures, and their properties

**REAL VARIABLES I**

**Fall 2026**

The preceding discussion of outer measure provides a size for every subset of $X$, but the Vitali counterexample shows that no countably additive, translation-invariant extension of interval length can be defined on every subset of $\mathbb{R}$. To obtain additivity, we must restrict the domain of a measure from $2^X$ to a carefully chosen collection of sets.

That collection must still support the standard operations consistens with measure properties: complements, countable unions, and therefore countable intersections. $\sigma$-algebras supply exactly this structure. Once the domain is in place, countable additivity yields the basic inequalities and continuity properties that make measures useful in later integration theory.

After studying these notes, you will be able to do the following.

- Distinguish rings, algebras, and $\sigma$-algebras and use their closure properties.
- Generate a $\sigma$-algebra and identify standard Borel generating families.
- Verify that a set function is a measure and derive its basic inequalities.
- Relate $\sigma$ additivity, finite superadditivity, and monotone continuity.
- Compute measures of monotone limits of sets and recognize why the finite-measure hypothesis in continuity from above is necessary.

## 10.1 Algebras, sigma-algebras, and generators

The domain of a measure must be stable under the operations used in countable decompositions. Rings and algebras isolate the finite operations; $\sigma$-algebras add countable unions.

**Definition 10.1.1** (Rings, algebras, and $\sigma$-algebras of sets)**.** Let $\mathcal{A} \subseteq 2^X$.

1. The collection $\mathcal{A}$ is a *ring of sets* if it is nonempty and, for every $A, B \in \mathcal{A}$,
$$A \cup B \in \mathcal{A} \qquad \text{and} \qquad A \setminus B \in \mathcal{A}.$$
2. The collection $\mathcal{A}$ is an *algebra of sets* if it is a ring of sets and $X \in \mathcal{A}$.
3. The collection $\mathcal{A}$ is a *$\sigma$-algebra* if $X \in \mathcal{A}$, if $X \setminus A \in \mathcal{A}$ for every $A \in \mathcal{A}$, and if
$$\bigcup_{n=1}^{\infty} A_n \in \mathcal{A}$$
for every sequence $(A_n \in \mathcal{A})_{n\in\mathbb{N}}$.

An algebra is equivalently a nonempty collection closed under complements relative to $X$ and finite unions. For the reverse implication, these two closure properties give

$$X = A \cup A^c, \qquad A \cap B = (A^c \cup B^c)^c, \qquad A \setminus B = A \cap B^c.$$

The countable-union requirement is the additional condition that distinguishes a $\sigma$-algebra from an algebra.

**Proposition 10.1.2** (Closure properties of set algebras). Let $\mathcal{A} \subseteq 2^X$.

1. If $\mathcal{A}$ is a ring of sets, then $\emptyset \in \mathcal{A}$, and $\mathcal{A}$ is closed under finite intersections.
2. If $\mathcal{A}$ is an algebra, then it is closed under finite unions, finite intersections, complements, and differences.
3. If $\mathcal{A}$ is a $\sigma$-algebra, then it is an algebra and is closed under countable intersections.

*Proof. (proves 10.1.2)*
The proof follows the hierarchy in the statement, adding the closure operations available at each level.

**Step 1: obtain the ring consequences.**
Choose $A \in \mathcal{A}$. Closure under differences gives $\emptyset = A \setminus A \in \mathcal{A}$. If $A, B \in \mathcal{A}$, then

$$A \cap B = A \setminus (A \setminus B),$$

so a ring is closed under intersections of two sets and hence under finite intersections.

**Step 2: obtain complements in an algebra.**
If $\mathcal{A}$ is an algebra, then $X \in \mathcal{A}$ and

$$A^c \coloneqq X \setminus A \in \mathcal{A}.$$

The remaining finite operations follow from the ring properties.

**Step 3: obtain countable intersections in a $\sigma$-algebra.**
Suppose that $\mathcal{A}$ is a $\sigma$-algebra. Closure under finite unions follows by adjoining empty sets to a finite family. If $(A_n \in \mathcal{A})_{n\in\mathbb{N}}$, De Morgan's law gives

$$\bigcap_{n=1}^{\infty} A_n = \left(\bigcup_{n=1}^{\infty} A_n^c\right)^c \in \mathcal{A}.$$

The preceding parts now show that $\mathcal{A}$ is an algebra. □

### 10.1.1 Examples of sigma-algebras

**Example 10.1.3** (Trivial and discrete $\sigma$-algebras). The collections

$$\{\emptyset, X\} \qquad \text{and} \qquad 2^X$$

are $\sigma$-algebras on $X$. They are called the *trivial* and *discrete* $\sigma$-algebras, respectively.

For either collection, the complement and countable-union axioms follow directly from the displayed list of sets. The trivial $\sigma$-algebra retains only the distinction between the empty set and the whole space; the discrete $\sigma$-algebra admits every subset.

**Example 10.1.4** ($\sigma$-algebra of a finite partition). Let $P_1, \dots, P_N$ be nonempty, pairwise disjoint subsets of $X$ such that

$$X = \bigcup_{j=1}^{N} P_j.$$

Then

$$\mathcal{A} \coloneqq \left\{ \bigcup_{j \in J} P_j \,:\, J \subseteq \{1, \dots, N\} \right\}$$

is a $\sigma$-algebra. Its elements are precisely the unions of cells of the partition.

Indeed, the empty union is $\emptyset$ and the union of all cells is $X$. The complement of the union indexed by $J$ is the union indexed by $\{1, \dots, N\} \setminus J$. A countable union of elements of $\mathcal{A}$ is the union of the cells appearing in at least one of them, and is therefore again in $\mathcal{A}$.

**Example 10.1.5** (Countable–cocountable $\sigma$-algebra)**.** Define

$$\mathcal{A}_{\mathrm{cc}} \coloneqq \{A \subseteq X \,:\, A \text{ is countable or } A^c \text{ is countable}\}.$$

Then $\mathcal{A}_{\mathrm{cc}}$ is a $\sigma$-algebra on $X$.

The sets $\emptyset$ and $X$ belong to $\mathcal{A}_{\mathrm{cc}}$, and the definition is preserved by complements. Let $(A_n \in \mathcal{A}_{\mathrm{cc}})_{n \in \mathbb{N}}$. If every $A_n$ is countable, then $\bigcup_n A_n$ is countable. If some $A_{n_0}$ is cocountable, then

$$\left( \bigcup_{n=1}^{\infty} A_n \right)^c \subseteq A_{n_0}^c,$$

so the union is cocountable. This verifies closure under countable unions.
If $X$ is countable, then $\mathcal{A}_{\mathrm{cc}} = 2^X$. If $X$ is uncountable, the collection can be strictly smaller than $2^X$: any set for which both it and its complement are uncountable is excluded.

### 10.1.2 Generated sigma-algebras

A prescribed family of sets need not itself be a $\sigma$-algebra. The next construction adds every set forced by the $\sigma$-algebra axioms and no others.

**Proposition 10.1.6** (Intersections of $\sigma$-algebras)**.** Let $I$ be a nonempty index set, and let $(\mathcal{A}_i)_{i \in I}$ be $\sigma$-algebras on $X$. Then

$$\bigcap_{i \in I} \mathcal{A}_i$$

is a $\sigma$-algebra on $X$.

*Proof. (proves 10.1.6)*
Every $\mathcal{A}_i$ contains $X$, so their intersection contains $X$. If $A$ belongs to every $\mathcal{A}_i$, then $A^c$ belongs to every $\mathcal{A}_i$, and therefore belongs to the intersection. If every $A_n$ belongs to the intersection, then each $\mathcal{A}_i$ contains every $A_n$ and hence contains $\bigcup_n A_n$. Thus the union belongs to the intersection. $\square$

**Definition 10.1.7** (Generated $\sigma$-algebra)**.** Let $\mathcal{G} \subseteq 2^X$. The *$\sigma$-algebra generated by $\mathcal{G}$* is

$$\sigma(\mathcal{G}) \coloneqq \bigcap \{\mathcal{A} \subseteq 2^X \,:\, \mathcal{A} \text{ is a } \sigma\text{-algebra on } X \text{ and } \mathcal{G} \subseteq \mathcal{A}\}.$$

The collection being intersected is nonempty because $2^X$ is a $\sigma$-algebra containing $\mathcal{G}$. By Proposition 10.1.6, $\sigma(\mathcal{G})$ is a $\sigma$-algebra. It contains $\mathcal{G}$, and it is contained in every $\sigma$-algebra that contains $\mathcal{G}$. These facts characterize $\sigma(\mathcal{G})$ as the smallest $\sigma$-algebra containing $\mathcal{G}$.

**Corollary 10.1.8** (Comparison of generators)**.** Let $\mathcal{G}, \mathcal{H} \subseteq 2^X$. If $\mathcal{G} \subseteq \sigma(\mathcal{H})$, then

$$\sigma(\mathcal{G}) \subseteq \sigma(\mathcal{H}).$$

Consequently, if both $\mathcal{G} \subseteq \sigma(\mathcal{H})$ and $\mathcal{H} \subseteq \sigma(\mathcal{G})$, then $\sigma(\mathcal{G}) = \sigma(\mathcal{H})$.

*Proof. (proves 10.1.8)*
The $\sigma$-algebra $\sigma(\mathcal{H})$ contains $\mathcal{G}$. The minimality in Definition 10.1.7 therefore gives $\sigma(\mathcal{G}) \subseteq \sigma(\mathcal{H})$. Apply the same argument in both directions for the final assertion. $\square$

Generated $\sigma$-algebras are usually handled through this minimality property. For example, to compare two generating families, it suffices to show that each member of one family belongs to the $\sigma$-algebra generated by the other. An explicit list of all elements of $\sigma(\mathcal{G})$ is generally unavailable because the axioms may be applied repeatedly to countable families.

### 10.1.3 Borel sigma-algebras

**Definition 10.1.9** (Borel $\sigma$-algebra)**.** Let $(X, \mathcal{T})$ be a topological space. The *Borel $\sigma$-algebra* of $X$ is

$$\mathcal{B}(X) \coloneqq \sigma(\mathcal{T}),$$

the $\sigma$-algebra generated by the open subsets of $X$. Its elements are the *Borel sets* of $X$.

On Euclidean space, a countable family of rectangles generates all Borel sets.

**Proposition 10.1.10** (Rational rectangles generate the Euclidean Borel sets)**.** Let $d \in \mathbb{N}$ with $d \geq 1$, and let

$$\mathcal{R}_{\mathbb{Q}} \coloneqq \left\{ \prod_{j=1}^{d} (a_j, b_j) \,:\, a_j, b_j \in \mathbb{Q},\ a_j < b_j \text{ for } 1 \leq j \leq d \right\}.$$

Then $\mathcal{R}_{\mathbb{Q}}$ is countable, is a base for the Euclidean topology on $\mathbb{R}^d$, and

$$\mathcal{B}(\mathbb{R}^d) = \sigma(\mathcal{R}_{\mathbb{Q}}).$$

*Proof. (proves 10.1.10)*
The proof establishes countability, the basis property, and then the generation statement in that order.

**Step 1: count the rational rectangles.**

Every rectangle in $\mathcal{R}_{\mathbb{Q}}$ is determined by an element of $\mathbb{Q}^{2d}$. Since a finite Cartesian product of countable sets is countable, $\mathcal{R}_{\mathbb{Q}}$ is countable.

**Step 2: place a rational rectangle inside every neighborhood.**

Let $U \subseteq \mathbb{R}^d$ be open and let $x \in U$. Choose $r > 0$ such that the Euclidean ball $B_r(x)$ is contained in $U$. Density of $\mathbb{Q}$ gives rational numbers $a_j, b_j$ satisfying

$$x_j - \frac{r}{\sqrt{d}} < a_j < x_j < b_j < x_j + \frac{r}{\sqrt{d}} \qquad (1 \leq j \leq d).$$

For

$$R = \prod_{j=1}^{d} (a_j, b_j),$$

we have $x \in R \subseteq B_r(x) \subseteq U$. Thus $\mathcal{R}_{\mathbb{Q}}$ is a base.

**Step 3: pass from the basis to the Borel $\sigma$-algebra.**

Every open set $U$ is the union of all rectangles $R \in \mathcal{R}_{\mathbb{Q}}$ satisfying $R \subseteq U$. This is a countable union because $\mathcal{R}_{\mathbb{Q}}$ is countable. Hence $U \in \sigma(\mathcal{R}_{\mathbb{Q}})$, which gives

$$\mathcal{B}(\mathbb{R}^d) \subseteq \sigma(\mathcal{R}_{\mathbb{Q}}).$$

Every rational rectangle is open, so the reverse inclusion follows from the definition of $\mathcal{B}(\mathbb{R}^d)$. □

The choice of open rational rectangles is convenient but not essential.

**Proposition 10.1.11** (Equivalent rectangle generators). Let $d \in \mathbb{N}$ with $d \geq 1$. Let $\mathcal{R}_{\mathrm{op}}$, $\mathcal{R}_{\mathrm{cl}}$, and $\mathcal{R}_{\mathrm{ho}}$ denote, respectively, the collections of bounded axis-parallel rectangles of the forms

$$\prod_{j=1}^{d}(a_j, b_j), \qquad \prod_{j=1}^{d}[a_j, b_j], \qquad \prod_{j=1}^{d}[a_j, b_j),$$

where $a_j < b_j$ are real numbers. Then

$$\sigma(\mathcal{R}_{\mathrm{op}}) = \sigma(\mathcal{R}_{\mathrm{cl}}) = \sigma(\mathcal{R}_{\mathrm{ho}}) = \mathcal{B}(\mathbb{R}^d).$$

The same conclusion holds if every endpoint is required to be rational.

*Proof. (proves 10.1.11)*
The proof first puts every generator inside the Borel $\sigma$-algebra and then reconstructs open rectangles from the other endpoint conventions.

**Step 1: obtain the inclusions into the Borel sets.**

Open rectangles are open, closed rectangles are closed, and half-open rectangles are finite intersections of open and closed coordinate half-spaces. Every rectangle in the three families is therefore Borel. This gives one inclusion for each generated $\sigma$-algebra.

**Step 2: reconstruct open rectangles.**

By Proposition 10.1.10, open rectangles generate $\mathcal{B}(\mathbb{R}^d)$. It remains to recover each open rectangle from closed or half-open rectangles. Fix

$$R = \prod_{j=1}^{d}(a_j, b_j),$$

and choose $n_0$ so large that $2/n_0 < \min_j(b_j - a_j)$. Then

$$R = \bigcup_{n=n_0}^{\infty} \prod_{j=1}^{d}[a_j + 1/n, b_j - 1/n].$$

Thus $R \in \sigma(\mathcal{R}_{\mathrm{cl}})$. After increasing $n_0$ if necessary so that $1/n_0 < \min_j(b_j - a_j)$, we also have

$$R = \bigcup_{n=n_0}^{\infty} \prod_{j=1}^{d}[a_j + 1/n, b_j),$$

so $R \in \sigma(\mathcal{R}_{\mathrm{ho}})$. The generator comparison in Corollary 10.1.8 proves the equalities.

**Step 3: retain rational endpoints.**

If all original endpoints are rational, then every approximating endpoint in the two displays is rational. Together with Proposition 10.1.10, this proves the rational-endpoint assertion. □

**Definition 10.1.12** (Measurable space). A *measurable space* is a pair $(X, \mathcal{A})$ in which $X$ is a set and $\mathcal{A}$ is a $\sigma$-algebra on $X$. The elements of $\mathcal{A}$ are called the *measurable sets* of the space.

The adjective "measurable" is always relative to the selected $\sigma$-algebra. The same underlying set $X$ may be equipped with different $\sigma$-algebras.

## 10.2 Measures and their basic inequalities

The $\sigma$-algebra supplies the domain. A measure assigns a nonnegative extended real number to every set in that domain and is additive along countable disjoint decompositions.

**Definition 10.2.1** (Measure and measure space). Let $(X, \mathcal{A})$ be a measurable space. A *measure* on $(X, \mathcal{A})$ is a function

$$\mu\colon \mathcal{A} \longrightarrow [0, +\infty]$$

such that

1. $\mu(\emptyset) = 0$;
2. for every pairwise disjoint sequence $(E_n \in \mathcal{A})_{n\in\mathbb{N}}$,

$$\mu\left(\bigcup_{n=1}^{\infty} E_n\right) = \sum_{n=1}^{\infty} \mu(E_n).$$

The second property is *countable additivity* or *$\sigma$-additivity*. A triple $(X, \mathcal{A}, \mu)$ is a *measure space*.

The additivity requirement concerns disjoint sets. Countable subadditivity for arbitrary measurable sets will follow from it.

### 10.2.1 Examples and size conditions

**Example 10.2.2** (Weighted counting measures). Let $w\colon X \to [0, +\infty]$. For $E \subseteq X$, define

$$\mu_w(E) \coloneqq \sum_{x\in E} w(x),$$

where the nonnegative sum is the supremum of its finite subsums. Then $\mu_w$ is a measure on $(X, 2^X)$.

The empty sum is zero. If $(E_n)_{n\in\mathbb{N}}$ is pairwise disjoint, then every $x \in \bigcup_n E_n$ occurs in exactly one of the sets. Reindexing the nonnegative sum gives

$$\mu_w\left(\bigcup_{n=1}^{\infty} E_n\right) = \sum_{x\in\bigcup_n E_n} w(x) = \sum_{n=1}^{\infty}\sum_{x\in E_n} w(x) = \sum_{n=1}^{\infty} \mu_w(E_n).$$

The definition by finite subsums makes this reindexing valid whether the value is finite or infinite. Taking $w(x) = 1$ for every $x$ gives *counting measure*:

$$\#E = \begin{cases} \text{the number of elements of } E, & E \text{ finite},\\ +\infty, & E \text{ infinite}.\end{cases}$$

**Example 10.2.3** (Dirac point mass). Let $(X, \mathcal{A})$ be a measurable space and fix $x_0 \in X$. The *Dirac measure at $x_0$* is

$$\delta_{x_0}(E) \coloneqq \begin{cases} 1, & x_0 \in E, \\ 0, & x_0 \notin E, \end{cases} \qquad E \in \mathcal{A}.$$

We have $\delta_{x_0}(\emptyset) = 0$. For a pairwise disjoint sequence, at most one set contains $x_0$. Both sides of the countable-additivity identity are therefore 1 if $x_0$ belongs to the union and 0 otherwise.

**Example 10.2.4** (Anticipated Lebesgue measure). For a positive integer $d$, we will later construct a $\sigma$-algebra $\mathcal{L}(\mathbb{R}^d)$ containing $\mathcal{B}(\mathbb{R}^d)$ and a measure

$$\lambda \colon \mathcal{L}(\mathbb{R}^d) \longrightarrow [0, +\infty]$$

such that $\lambda(R)$ is the Euclidean volume of every axis-parallel rectangle $R$. This measure is Lebesgue measure. Its construction from Lebesgue outer measure is postponed.

**Definition 10.2.5** (Finite, probability, and $\sigma$-finite measures). Let $(X, \mathcal{A}, \mu)$ be a measure space.

1. The measure $\mu$ is *finite* if $\mu(X) < +\infty$.
2. The measure $\mu$ is a *probability measure* if $\mu(X) = 1$.
3. The measure $\mu$ is *$\sigma$-finite* if there is a sequence $(X_n \in \mathcal{A})_{n\in\mathbb{N}}$ such that

$$X = \bigcup_{n=1}^{\infty} X_n \qquad \text{and} \qquad \mu(X_n) < +\infty \quad \text{for every } n.$$

Every probability measure is finite, and every finite measure is $\sigma$-finite by taking $X_n = X$. The converses fail. Counting measure on $\mathbb{N}$ is $\sigma$-finite because $\mathbb{N}$ is a countable union of singletons, but it is not finite. Counting measure on an uncountable set $X$ is not $\sigma$-finite: every set of finite counting measure is finite, while a countable union of finite sets is countable. The Dirac measure $\delta_{x_0}$ is a probability measure. On $X = \{1, 2, 3, \ldots\}$, the weighted counting measure with $w(n) = 2^{-n}$ is also a probability measure.
Conditional on the construction in Example 10.2.4, Lebesgue measure on $\mathbb{R}^d$ is $\sigma$-finite because the boxes $[-n, n]^d$ have finite measure and cover $\mathbb{R}^d$. It is not finite: monotonicity and the rectangle formula give

$$\lambda(\mathbb{R}^d) \geq \lambda([-n, n]^d) = (2n)^d$$

for every $n \geq 1$.

### 10.2.2 Consequences of countable additivity

**Proposition 10.2.6** (Finite additivity, monotonicity, and differences). Let $(X, \mathcal{A}, \mu)$ be a measure space.

1. If $E_1, \ldots, E_N \in \mathcal{A}$ are pairwise disjoint, then

$$\mu\left(\bigcup_{n=1}^{N} E_n\right) = \sum_{n=1}^{N} \mu(E_n).$$

2. If $A, B \in \mathcal{A}$ and $A \subseteq B$, then $\mu(A) \leq \mu(B)$.
3. If $A, B \in \mathcal{A}$, $A \subseteq B$, and $\mu(A) < +\infty$, then

$$\mu(B \setminus A) = \mu(B) - \mu(A).$$

*Proof.* *(proves 10.2.6)*
The proof obtains finite additivity directly and then applies it to the disjoint decomposition of a larger set.

**Step 1: reduce finite additivity to countable additivity.**

For finite additivity, apply countable additivity to the sequence $E_1, \dots, E_N, \emptyset, \emptyset, \dots$.

**Step 2: decompose a set difference.**

If $A \subseteq B$, then $B = A \mathbin{\dot\cup} (B \setminus A)$, where the dot records a disjoint union. Finite additivity gives

$$\mu(B) = \mu(A) + \mu(B \setminus A) \geq \mu(A).$$

If $\mu(A) < +\infty$, subtraction of the finite quantity $\mu(A)$ is defined and gives the difference formula. ☐

**Warning** (Cancellation at infinity)**.** The difference formula cannot be obtained by formal cancellation when $\mu(A) = +\infty$. For counting measure on $X = \mathbb{N} \cup \{p\}$, take $A = \mathbb{N}$ and $B = X$. Then $\mu(B \setminus A) = 1$, while the expression $\mu(B) - \mu(A) = +\infty - (+\infty)$ is undefined.

**Proposition 10.2.7** (Countable superadditivity and subadditivity)**.** Let $(X, \mathcal{A}, \mu)$ be a measure space.

1. If $E \in \mathcal{A}$ and $(E_n \in \mathcal{A})_{n\in\mathbb{N}}$ is pairwise disjoint with $E_n \subseteq E$ for every $n$, then

$$\sum_{n=1}^{\infty} \mu(E_n) \leq \mu(E).$$

2. For every sequence $(A_n \in \mathcal{A})_{n\in\mathbb{N}}$,

$$\mu\left(\bigcup_{n=1}^{\infty} A_n\right) \leq \sum_{n=1}^{\infty} \mu(A_n).$$

*Proof.* *(proves 10.2.7)*
The first assertion uses additivity on an already disjoint family. The second turns an arbitrary family into a disjoint one.

**Step 1: compare a disjoint subfamily with its container.**

For the first assertion, countable additivity and monotonicity give

$$\sum_{n=1}^{\infty} \mu(E_n) = \mu\left(\bigcup_{n=1}^{\infty} E_n\right) \leq \mu(E).$$

**Step 2: disjointify an arbitrary sequence.**

For subadditivity, disjointify the sequence by setting

$$F_1 = A_1, \qquad F_n = A_n \setminus \bigcup_{k=1}^{n-1} A_k \quad (n \geq 2).$$

The sets $F_n$ are measurable and pairwise disjoint,

$$\bigcup_{n=1}^{\infty} F_n = \bigcup_{n=1}^{\infty} A_n, \qquad F_n \subseteq A_n.$$

Countable additivity and monotonicity yield

$$\mu\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} \mu(F_n) \leq \sum_{n=1}^{\infty} \mu(A_n).$$

□

The two-set form of inclusion–exclusion can be written without any subtraction:

$$\mu(A \cup B) + \mu(A \cap B) = \mu(A) + \mu(B).$$

Indeed, after decomposing $A \cup B$ into the three disjoint sets $A \setminus B$, $A \cap B$, and $B \setminus A$, both sides equal

$$\mu(A \setminus B) + \mu(B \setminus A) + 2\mu(A \cap B).$$

The identity remains meaningful when one or both sides are infinite.

**Corollary 10.2.8** (Finite inclusion–exclusion)**.** Let $A_1, \dots, A_N \in \mathcal{A}$, and assume $\mu(A_j) < +\infty$ for every $j$. Then every term below is finite and

$$\mu\left(\bigcup_{j=1}^{N} A_j\right) = \sum_{\emptyset \neq J \subseteq \{1,\dots,N\}} (-1)^{|J|+1} \mu\left(\bigcap_{j \in J} A_j\right).$$

*Proof. (proves 10.2.8)*
Every nonempty intersection in the formula is contained in one of the sets $A_j$, so its measure is finite by monotonicity. The formula for $N = 2$ follows by subtracting the finite value $\mu(A_1 \cap A_2)$ from the two-set identity. Induction on $N$, applying the two-set formula to $\bigcup_{j=1}^{N-1} A_j$ and $A_N$, gives the displayed identity. □

## 10.3 Continuity of measures and Borel sets

Countable additivity determines the measure of a monotone union or intersection from the measures of its terms. The required set operations are recorded first.

### 10.3.1 Limits of sets

**Definition 10.3.1** (Increasing and decreasing sequences of sets)**.** Let $(E_n \subseteq X)_{n \in \mathbb{N}}$.

1. We write $E_n \uparrow E$ if $E_n \subseteq E_{n+1}$ for every $n$ and

$$E = \bigcup_{n=1}^{\infty} E_n.$$

2. We write $E_n \downarrow E$ if $E_{n+1} \subseteq E_n$ for every $n$ and

$$E = \bigcap_{n=1}^{\infty} E_n.$$

**Definition 10.3.2** (Lower and upper set limits)**.** For a sequence $(E_n \subseteq X)_{n\in\mathbb{N}}$, define

$$\liminf_{n\to\infty} E_n \coloneqq \bigcup_{N=1}^{\infty} \bigcap_{n=N}^{\infty} E_n$$

and

$$\limsup_{n\to\infty} E_n \coloneqq \bigcap_{N=1}^{\infty} \bigcup_{n=N}^{\infty} E_n.$$

A point belongs to $\liminf_n E_n$ if it belongs to every $E_n$ from some index onward. A point belongs to $\limsup_n E_n$ if it belongs to infinitely many of the sets $E_n$. Consequently,

$$\liminf_{n\to\infty} E_n \subseteq \limsup_{n\to\infty} E_n.$$

**Proposition 10.3.3** (Measurability of set limits)**.** Let $(X, \mathcal{A})$ be a measurable space and let $(E_n \in \mathcal{A})_{n\in\mathbb{N}}$. Then

$$\bigcup_{n=1}^{\infty} E_n, \quad \bigcap_{n=1}^{\infty} E_n, \quad \liminf_{n\to\infty} E_n, \quad \text{and} \quad \limsup_{n\to\infty} E_n$$

all belong to $\mathcal{A}$.

*Proof. (proves 10.3.3)*
The $\sigma$-algebra axioms and De Morgan's laws give closure under countable unions and countable intersections. Each lower or upper set limit in Definition 10.3.2 is obtained by applying these two operations twice. □

### 10.3.2 Continuity from below and above

**Theorem 10.3.4** (Continuity from below)**.** Let $(X, \mathcal{A}, \mu)$ be a measure space. If $(E_n \in \mathcal{A})_{n\in\mathbb{N}}$ and $E_n \uparrow E$, then

$$\mu(E_n) \uparrow \mu(E),$$

that is,

$$\lim_{n\to\infty} \mu(E_n) = \mu(E).$$

No finiteness hypothesis is required.

*Proof. (proves 10.3.4)*
Define the disjoint increments

$$D_1 = E_1, \qquad D_n = E_n \setminus E_{n-1} \quad (n \geq 2).$$

They are measurable and pairwise disjoint. Moreover,

$$E_N = \bigcup_{n=1}^{N} D_n \qquad \text{and} \qquad E = \bigcup_{n=1}^{\infty} D_n.$$

Finite and countable additivity therefore give

$$\mu(E_N) = \sum_{n=1}^{N} \mu(D_n) \quad \text{and} \quad \mu(E) = \sum_{n=1}^{\infty} \mu(D_n).$$

The partial sums of a nonnegative series increase to the series, so $\mu(E_N) \uparrow \mu(E)$. □

**Theorem 10.3.5** (Continuity from above). Let $(X, \mathcal{A}, \mu)$ be a measure space. If $(E_n \in \mathcal{A})_{n\in\mathbb{N}}$, $E_n \downarrow E$, and $\mu(E_1) < +\infty$, then

$$\mu(E_n) \downarrow \mu(E).$$

*Proof. (proves 10.3.5)*
Set

$$F_n = E_1 \setminus E_n.$$

Then $F_n \uparrow E_1 \setminus E$. By Theorem 10.3.4,

$$\mu(F_n) \uparrow \mu(E_1 \setminus E).$$

The sets $E_n$ and $F_n$ form a disjoint decomposition of $E_1$, as do $E$ and $E_1 \setminus E$. Since $\mu(E_1) < +\infty$, all terms are finite and

$$\mu(E_n) = \mu(E_1) - \mu(F_n).$$

Taking limits gives

$$\lim_{n\to\infty} \mu(E_n) = \mu(E_1) - \mu(E_1 \setminus E) = \mu(E).$$

□

**Example 10.3.6** (Failure of continuity from above without finite measure). Let $X = \mathbb{N}$, let $\mathcal{A} = \mathcal{P}(\mathbb{N})$, and let $\mu$ be counting measure. For

$$E_n = \{n, n+1, n+2, ...\},$$

we have $E_n \downarrow \emptyset$, but

$$\mu(E_n) = +\infty \quad \text{for every } n, \qquad \mu(\emptyset) = 0.$$

Thus continuity from above can fail when $\mu(E_1) = +\infty$.

Continuity from below also characterizes countable additivity once finite additivity is known.

**Proposition 10.3.7** (Countable additivity from continuity below). Let $(X, \mathcal{A})$ be a measurable space, and let

$$\nu\colon \mathcal{A} \longrightarrow [0, +\infty]$$

satisfy $\nu(\emptyset) = 0$ and finite additivity on disjoint measurable sets. Then $\nu$ is a measure if and only if it is continuous from below: whenever $(E_n \in \mathcal{A})_{n\in\mathbb{N}}$ and $E_n \uparrow E$,

$$\nu(E_n) \longrightarrow \nu(E).$$

*Proof. (proves 10.3.7)*
The forward direction is the continuity theorem; the reverse direction applies continuity to finite partial unions.

**Step 1: obtain continuity from a measure.**

If $\nu$ is a measure, continuity from below is Theorem 10.3.4.

**Step 2: recover countable additivity from continuity.**

Conversely, suppose that $\nu$ is continuous from below, and let $(A_n \in \mathcal{A})_{n\in\mathbb{N}}$ be pairwise disjoint. Define

$$S_N = \bigcup_{n=1}^{N} A_n, \qquad S = \bigcup_{n=1}^{\infty} A_n.$$

Then $S_N \uparrow S$. Finite additivity and continuity from below give

$$\nu(S) = \lim_{N\to\infty} \nu(S_N) = \lim_{N\to\infty} \sum_{n=1}^{N} \nu(A_n) = \sum_{n=1}^{\infty} \nu(A_n).$$

Thus $\nu$ is countably additive. $\square$

### 10.3.3 The Cantor set through continuity from above

The following computation is conditional on the later construction of Lebesgue measure. Its purpose here is to apply Theorem 10.3.5 to a decreasing geometric construction.

**Example 10.3.8** (Lebesgue measure of the Cantor set)**.** Assume that Lebesgue measure $\lambda$ has been constructed on a $\sigma$-algebra containing the closed subsets of $\mathbb{R}$ and that $\lambda([a, b]) = b - a$ for finite $a < b$. Let $C_0 = [0, 1]$. Given $C_n$, form $C_{n+1}$ by removing the open middle third of every interval component of $C_n$. Let

$$C = \bigcap_{n=0}^{\infty} C_n$$

be the middle-thirds Cantor set. Then $\lambda(C) = 0$.

At stage $n$, the set $C_n$ is a disjoint union of $2^n$ closed intervals, each of length $3^{-n}$. Finite additivity and agreement with interval length give

$$\lambda(C_n) = 2^n 3^{-n} = \left(\frac{2}{3}\right)^n.$$

The sets decrease to $C$, and $\lambda(C_0) = 1 < +\infty$. Continuity from above therefore gives

$$\lambda(C) = \lim_{n\to\infty} \lambda(C_n) = \lim_{n\to\infty} \left(\frac{2}{3}\right)^n = 0.$$

### 10.3.4 Borel sets obtained from sequences

If every $E_n$ is Borel in a topological space, Proposition 10.3.3 shows that $\liminf_n E_n$ and $\limsup_n E_n$ are Borel. The formula for $\limsup_n E_n$ gives the Borel set of points that lie in infinitely many of the sets. For a sequence of points, its subsequential limit set has a closed-set description.

**Proposition 10.3.9** (Limit points of a sequence form a closed set)**.** Let $(x_n \in \mathbb{R}^d)_{n\in\mathbb{N}}$, and let

$$L \coloneqq \left\{ x \in \mathbb{R}^d : x_{n_k} \longrightarrow x \text{ for some strictly increasing sequence } (n_k)_{k\in\mathbb{N}} \right\}.$$

Then

$$L = \bigcap_{N=1}^{\infty} \overline{\{x_n : n \geq N\}}.$$

Consequently, $L$ is closed and hence Borel.

*Proof. (proves 10.3.9)*
The two inclusions translate between subsequences and membership in every tail closure.

**Step 1: place each subsequential limit in every tail closure.**

Suppose $x_{n_k} \to x$. For every $N$, all sufficiently late indices $n_k$ satisfy $n_k \geq N$. Every neighborhood of $x$ therefore intersects $\{x_n : n \geq N\}$, so

$$x \in \overline{\{x_n : n \geq N\}}.$$

This proves the first inclusion.

**Step 2: choose a convergent subsequence from the tail closures.**

Conversely, suppose that $x$ belongs to every tail closure. Choose $n_1 \geq 1$ with $|x_{n_1} - x| < 1$. After $n_k$ has been chosen, membership in the closure of the tail beginning at $n_k + 1$ gives an index $n_{k+1} > n_k$ such that

$$|x_{n_{k+1}} - x| < \frac{1}{k+1}.$$

Thus $x_{n_k} \to x$, so $x \in L$. Each tail closure is closed, and a countable intersection of closed sets is closed. $\square$